



Fig. 3: Reference architecture of a wireless multimedia sensor network: (a) single-tier flat homogeneous sensors, distributed processing, centralised storage; (b) single-tier clustered, heterogeneous sensors, centralised processing, centralised storage; and (c) multi-tier, heterogeneous sensors, distributed processing, distributed storage (Courtesy: Akyildiz et al.: Wireless multimedia sensor networks: applications and test beds)

All cluster members send data to the cluster head, and hence the cluster head has the highest priority to allocate the channel for further communication. For any cluster member, inter-cluster and intra-cluster communication takes place via the cluster head.

Node with the highest energy is selected as cluster head as it consumes more energy. Special algorithms and protocols (for example, LEACH) elect the cluster head on the basis of remaining battery life (RBL) of all nodes within the cluster. A cluster that includes a variety of sensor nodes is called a heterogeneous sensor network (Fig. 3b).

A multi-tier architecture is arranged in hierarchy or multiple layers. As shown in Fig. 3(c), one cluster has formed of scalar nodes and the other using video sensors. Both clusters forward the data to the sink through high-end video sensors. The sink is connected to the Internet (IP based network) via a wireless gateway. This way the user can monitor and control targeted areas from a remote location.

Challenges and opportunities

High data rates. WMSNs require high data rates as data is in terms of

live video streaming, audio and still images. Achieving high data rates (in terms of Mbps) on available narrow channels is a challenging task. Recently, researchers started to implement ultra-wide band (UWB) technology on sensor nodes for higher data rates.

Signal detection and estimation. Signal detection, estimation, filtering, data gathering and wireless channel separation are still open research challenges for the implementation of WMSNs. Researchers have proposed OFDMs (frequency-division multiplexing schemes) to increase data rates and remove interference between two wireless channels.

Reliability. High information assurance is expected in WMSNs, and for that effective error detection, correction code with accuracy, robustness, resiliency and retransmission policy need to be found out.

Energy efficiency. Nodes have limited battery and, for most applications, charging and replacement of the battery is not possible. Nodes need to serve for many years. To reduce power consumption, researchers have already started designing and implementing different types of MAC protocols [real-time independent channels (RICH)

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